Autaptic inhibition's role in continual learning

Jens Friedel
School of Engineering Mathematics and Technology
University of Bristol
Bristol, United Kingdom
jens.friedel@bristol.ac.uk

Conor Houghton
School of Engineering Mathematics and Technology
University of Bristol
Bristol, United Kingdom
conor.houghton@bristol.ac.uk

Gabriel Leivas Oliveira
School of Engineering Mathematics and Technology
University of Bristol
Bristol, United Kingdom
gabriel.leivasoliveira@bristol.ac.uk

***Abstract*—TODO this is some sentence talking about how future AI and continual learning are important and causally related. How can one design a performant, memory-efficient continual learning framework that operates within a network's fixed capacity? We hypothesize that local capacity constraints limit long-term learning performance in sparse networks, and how distributing learning schemes could circumvent this problem. Inspired by biological networks (TODO: nec. start of sentence?)), we then propose autaptic inhibition for artificial intelligence (ai2) as a mechanism to motivate networks to distribute representations across many neurons. Such ai2 systems would outperform their conventionally sparse counterparts by balancing knowledge retaining and information redistribution flexibly.**

***Index Terms*—Collective dynamics, Game theory, Network formation games, Network theory (graphs)**

I. Introduction

II. Methods

III. Results and Discussion

IV. Conclusion

Network dynamics plays an ever-important role in describing the modern world, with connective plasticity and local optimizations enabling the emergence of global order. Since this framework can describe a vast variety of phenomena, researchers employed numerous approaches — often multi-disciplinary ones — to analyse these systems. Specifically, networks of human-like agents have been subject of studies ranging from mostly theoretical analysis to phenomenological social science works [1], [2].

When agent i sponsors an edge to j, we set $a_{ij} = 1$, and $a_{ij} = 0$ otherwise, so that

$$c_i^{\mathrm{NFG}}(a_i) = c_i^d(G) + \alpha_a \|a_i\| \tag{1}$$

returns the cost an agent incurs, with $\|\cdot\|$ being the L1 norm.

TABLE II
Total costs of ordered chains depend only on outermost opinions only, while agent utilities may differ significantly.

(x_1, x_2, x_3)	c_1^x	c_2^x	c_3^x
$(0, 0, 4)$	0	2	4
$(0, 2, 4)$	2	2	2
$(0, 1, 4)$	1	2	3

TABLE I
List of variables used in Co-Formation Games

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, ..., n\}$
Set of i's neighbours	K_i	$\mathbb{N}^{\|k_i\|}$
Network graph	G	$\{0,1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Opinion profile	x	$\mathbb{R}^n$
Action	a_i	$\{0,1\}^n$
Action profile	a	$\{0,1\}^{n \times n}$
Distance costs	c^d	$\mathbb{Z}^-$
Pairwise distance	d_{ij}	$\mathbb{N}$
Sponsorship costs	c^a	$]-\infty, 0[$
Costs per edge	α_a	$]0, \infty[$
Opinion costs	c^x	$]-\infty, 0[$
Opinion range	α_x	$\mathbb{R}^+$
Set of i's neighbours	K_i	$\mathbb{N}$
Communication rate	r_x	$[0, \infty[$
Action rate	r_a	$\{0, 1\}$
Convergence threshold	ε	$]0, \alpha_a]$

Lastly, this subsection lists some elementary conjectures of how chains and stars optimize costs given a certain opinion profile. As the proofs took place while trying to work out more general results, they are denoted as lemmata.

A. Lemma 1 – Optimal costs of chains

For n-node chains in a Co-Formation Game, the optimal total opinion costs, given an opinion set x reaching from 0 to α_x, are

1) $c_0^{\mathrm{opt}} = 0$ *if all x_i are identical,*
2) $c_{\mathrm{outside}}^{\mathrm{opt}} = \frac{3}{2}\alpha_x$ *if all but one x_i being identical, and*
3) $c_{\mathrm{inside}}^{\mathrm{opt}} = \alpha_x$ *otherwise.*

a) *Proof:*

Case 1 is trivial. Case 2 is achieved by placing the outlying opinion at the end of the chain, so that only one node suffers from the disagreement costs, which minimizes opinion costs c^x, as can be seen in (1). Case 3 is achieved by ordering opinions in the chain: $0 = x_1 \leq x_2 \leq ... \leq x_n = \alpha_x$. Each node i's opinion costs are

Initialize a, x
$\Delta c = 2\varepsilon$
while $(t < t_{\max} \wedge \varepsilon < \Delta c)$
  if $\mathrm{mod}(t, r_x) = 0$ **then**
    select random node i
    $x_i \leftarrow$ get opinion update x_i^u
  end if
  if $\mathrm{mod}(t, r_a) = 0$ **then**
    select random node i
    $a_i \leftarrow$ get action update a_i^u
  end if
  t += 1
  $c(t) \leftarrow$ get new costs c_t
  $\Delta c = c_{t-1} - c(t)$
end while

Algorithm 1. Schematic CFG algorithm pseudocode. Note this algorithm expects node i to have a cost-reducing action step available, whereas the actual implementation looks through nodes until it found one such step or probed all agents.

$$c_i^x = \frac{x_i - x_{i-1} + x_{i+1} - x_i}{2} = \frac{x_{i+1} - x_{i-1}}{2}, \qquad (2)$$

so that the total opinion costs become

$$c^x = \sum_{i=1}^{n} c_i^x = \sum_{i=1}^{n} \frac{x_{i+1} - x_{i-1}}{2} = x_n - x_1 = \alpha_x \qquad (3)$$

■.



G	c^d	c^a/α_a	c^x/α_x
top chain	8	2	1.5
bottom chain	8	2	3
triangle	6	3	2

Fig. 1. Depending on opinion placement, edge costs, and opinion range, costs motivate nodes to drop or add new edges.



G	c^d	c^a/α_a	c^x/α_x
top star	18	3	1.33
top chain	20	3	1.5
bottom chain	20	3	1
bottom star	18	3	2.66

Fig. 2. Chains can offer optimal costs, outdoing star graphs for certain opinion profiles and parameter choices by profiting from shrinking opinion costs when disagreement happens inside the chain.



Fig. 3. CFGs' number of steps until convergence, when starting with sparse topologies, scale with $n!$ when low edge costs favour dense graphs.

References

[1] P. Sobkowicz, "Whither Now, Opinion Modelers?," in *Frontiers of Physics*, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:226981914

[2] A. Flache *et al.*, "Models of Social Influence: Towards the Next Frontiers," *J. Artif. Soc. Soc. Simul.*, vol. 20, 2017, [Online]. Available: https://api.semanticscholar.org/CorpusID:3433103



Fig. 4. CFG runs converge hit the convergence threshold before opinion convergence can take place.